

How Drug Characteristics Affect User Satisfaction

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Problem Statement: This project examines what affects user satisfaction with drugs, including factors such as effectiveness, ease of use, price, form, and the medical condition treated. Satisfaction matters because it influences how well patients follow treatments and how they feel about their care. By examining user ratings alongside drug availability, these insights identify the most influential features and support more informed choices for patients, clinicians, and pharmaceutical developers.

Method:

- **Dataset:** The dataset, from Kaggle's Drug Performance Evaluation, contains information on drug characteristics and user satisfaction, allowing analysis of which features influence satisfaction.
- **Data Visualization:** I created the dashboard using Python with Matplotlib and Seaborn, cleaning the data and selecting the necessary columns. Bar charts show satisfaction by form and the number of available treatments for each condition, heatmaps display feature correlations, and a scatterplot examines ease of use versus effectiveness, highlighting the factors that most influence user satisfaction.

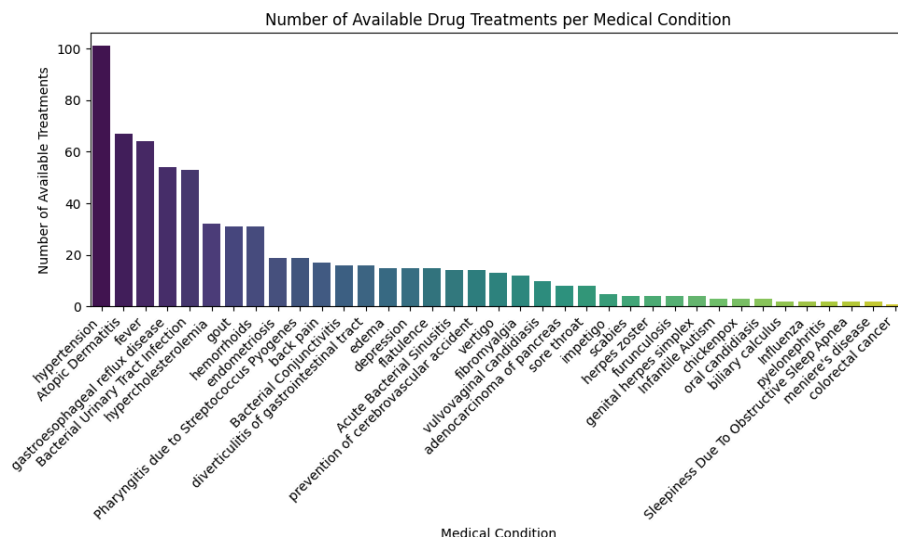


Figure 1: This bar chart shows the variation in the number of available drugs across different medical conditions from greatest to least. Hypertension stands out with the highest number of drugs, around 100, reflecting that it is a common condition with many

treatment options. In contrast, colorectal cancer has the fewest drugs available, with almost none listed, highlighting the limited treatment options for this condition. Other conditions fall in between, showing varying levels of drug availability. The chart emphasizes that the range of treatment options can differ greatly depending on the medical condition.

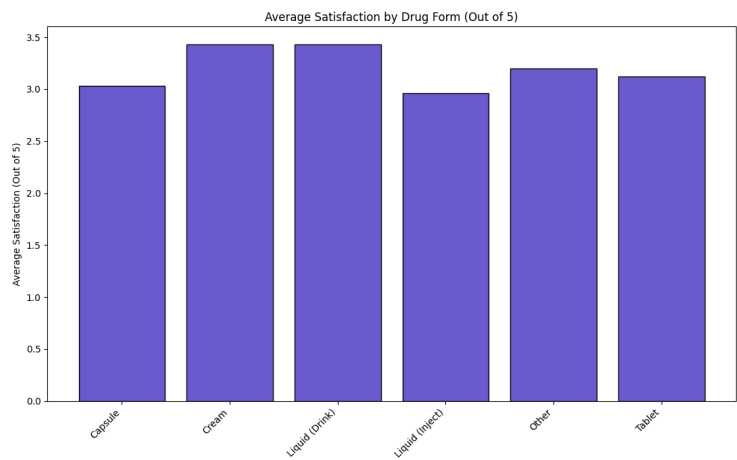


Figure 2: This bar chart compares the average satisfaction ratings of different drug forms on a scale from 1 to 5. The results indicate that liquid (drink) and cream forms have the highest average satisfaction, both around 3.5, suggesting that users generally find these forms easier or more pleasant to use. In contrast, liquid (inject) has the lowest average satisfaction, around 3, which may reflect that users find this form less convenient or comfortable. Tablet and capsule forms fall in between, showing moderate satisfaction levels. These differences suggest that the form of a medicine can influence how positively users perceive their experience with it.

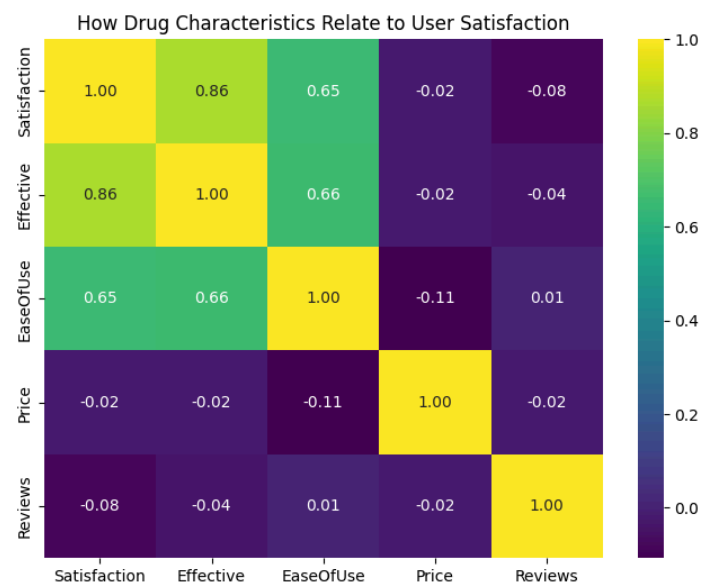


Figure 3: This heatmap illustrates how key drug characteristics relate to overall user satisfaction. Effectiveness shows the strongest positive relationship with correlation value 0.86, meaning users rate a medication higher when it works well for them. Ease of use also has a moderate positive association with correlation value 0.66, suggesting that simpler medications tend to receive better ratings. In contrast, price and number of reviews show negative relationships with satisfaction, indicating that higher cost and a greater number of reviews may actually reduce how positively users evaluate their medication experience.

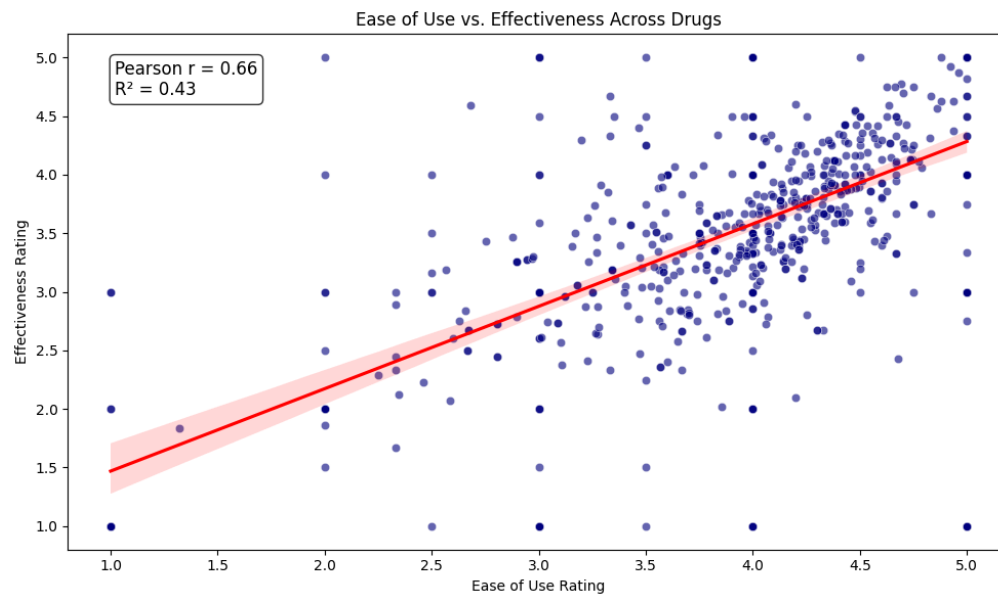


Figure 4: This scatterplot shows the relationship between a drug's ease of use and its perceived effectiveness. The relationship is positive: drugs with higher ease-of-use ratings generally receive higher effectiveness ratings. The points illustrate how effectiveness varies across ease-of-use levels and highlight clusters of drugs with similar ratings as well as outliers, especially with higher ease-of-use and effectiveness ratings. Statistically, the Pearson correlation is 0.66 and the R^2 value is 0.43, indicating a moderate positive relationship. This plot shows that medicines that are easier to use tend to be rated as more effective.

Results: The analysis showed that effectiveness and ease of use are the main factors affecting user satisfaction, with easier-to-use drugs generally rated as more effective. There is a moderate to strong positive relationship between ease of use and perceived effectiveness, with a Pearson correlation of 0.66 and an R^2 of 0.43. The drug form also affects satisfaction, with drinkable liquid and cream forms rated higher and injectable liquids lower. Drug availability varies by condition, helping patients choose medicines, guiding providers, and supporting drug makers in improving user experience.